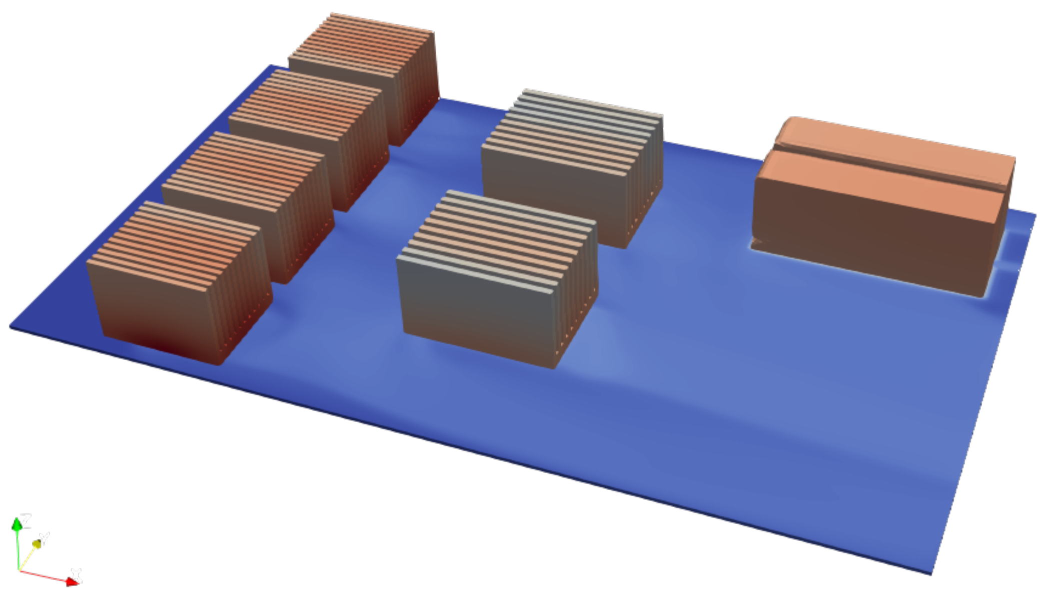
Energy-Efficient Electronics and Design Automation ($\underline{E^3DA}$) Lab

Engineered Multi-Physical Interactions & Reliability Evaluation (EMPIRE) Lab

PowerSynth 2 MOSTCOOL User Manual



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1 Introduction

1.1 PowerSynth Introduction

PowerSynth is an electronic design automation (EDA) tool that can synthesize and optimize multi-chip power module (MCPM) layouts with significantly faster than any commercial tools. PowerSynth currently performs multi-objective optimization to produce Pareto-front solutions to the proper placement of power semiconductor device die and the routing of metal traces on ceramic substrates. The tool accounts for temperature distributions and electrical parasitics as a function of the layout geometries that it considers. This tool has been hardware-validated. Continued research on this project will further elaborate the capabilities by extending the work to greater fidelity in thermal, electrical, and mechanical domains.

1.2 About - PowerSynth2-MOSTCOOL

This version (MOSTCOOL) offers a command line interface and a graphical user interface (GUI) on Linux machines for the design of the server board. In this work, server board design is developed based on thermal management. In this version, OpenFOAM API is developed which is a CFD free software. currently, air cooling technology is considered. Also, heatsink design can be defined in this version.

2 Using PowerSynth2 MOSTCOOL

2.1 Installation and Usage

PowerSynth2 MOSTCOOL can be installed on Linux machines. PowerSynth2 MOSTCOOL requires OpenFOAM software to run the thermal model. More details on the installation of OpenFOAM can be found in

`https://develop.openfoam.com/Development/openfoam/-/wikis/precompiled`.

It is recommended to install the OpenFOAM 22 or above.

After installation of OpenFOAM the user should install PowerSynth2-MOSTCOOL software. For installing the PowerSynth you can find information in the README.md within the pkg folder.

After installation of PowerSynth and OpenFOAM, to run PowerSynth2-MOSTCOOL software, in the terminal first set the path to include the bin folder, then use command PS2MostCool:

```
export PATH=`realpath bin`:$PATH
PS2MostCool
```

2.2 PowerSynth2-MOSTCOOL GUI Introduction

Upon running PowerSynth2-MOSTCOOL (PS2MostCool) executable, the following main window (shown in Figure 1) will appear. The window has four tabs: (a) **Main** tab to define Case study, OpenFOAM parameters, Server Parameters, and Reliability Settings ,(b) **CPUs** tab to defines CPUs, (c) **GPUs** tab to defines GPUs, and (d) **PSUs** tab to defines PSUs.

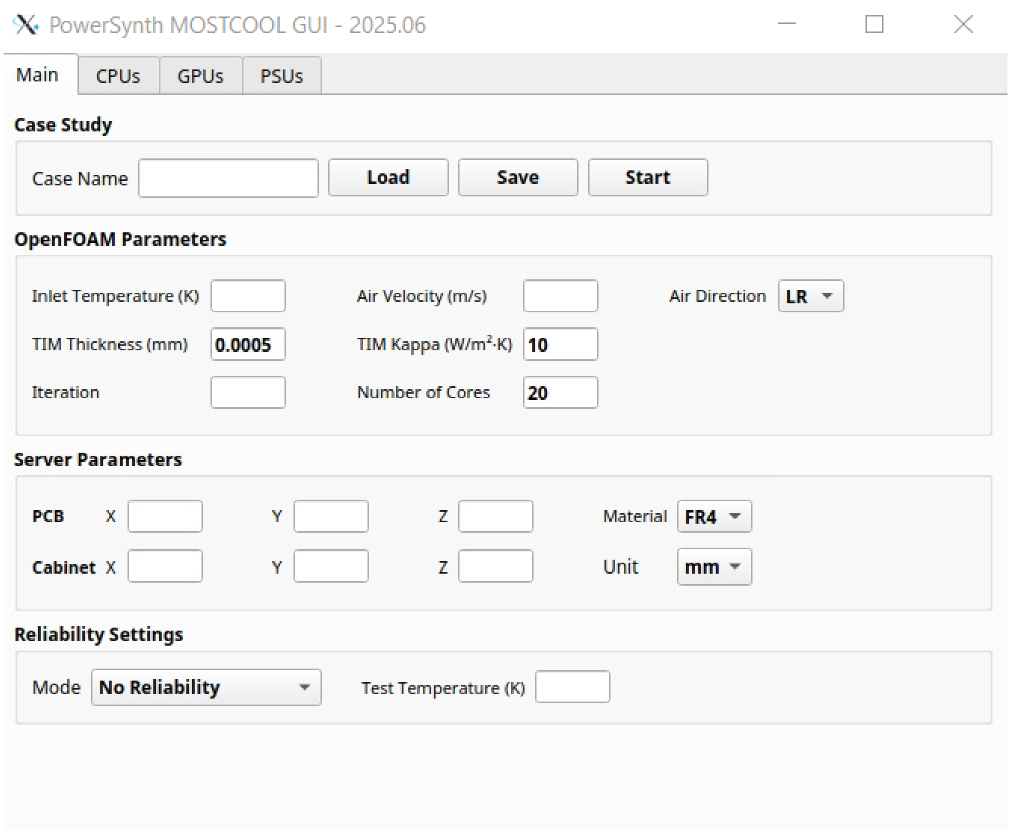


Figure 1: PowerSynth2-MOSTCOOL main window

(a) **Main:** In the Main tab users should define necessary initial parameters.

- Case study section includes: Case Name, Load button to load a pre-define cade study YAML file, Save button to save as a new case, and Start button to run the software for simulation. It is noted that, the input format is based on YAML and will be explained in the next section.

- Section OpenFOAM Parameters include the boundary conditions and solver settings. Inlet Temperature, Air Velocity and Air Direction, Thermal Interface Material (TIM), Iteration to simulation, and Number of Cores to parallel simulation.

- Section Server Parameters includes PCB dimensions and material and Cabinet dimensions. Also, the user can define the unit for representation.

- Section Reliability Settings includes mode consisting of no reliability, simple reliability (build-in model), and advanced reliability (API to other models), and test temperature.

(b) **CPUs:** In this tab users should define CPUs parameters. initially the first table shows the default CPU type. there are two type of CPU as a default. The users can also define their own CPU using the Add CPU Type button. Also, using the Remove CPU Type button can remove the types. After define the type of CPU, the number of CPUs that they want to use in the server board should be defined.

Based on the number automatically the second table is created. The second table includes chip and heatsink parameters. chip: location of CPU on the board (X, Y) and type that already is defined in first table, heatsink: dimensions (X,Y,Z, Tb), nFins(number of fins), fin width, and material. Figure 2 is shown the initial state and final state of the CPUs tab.

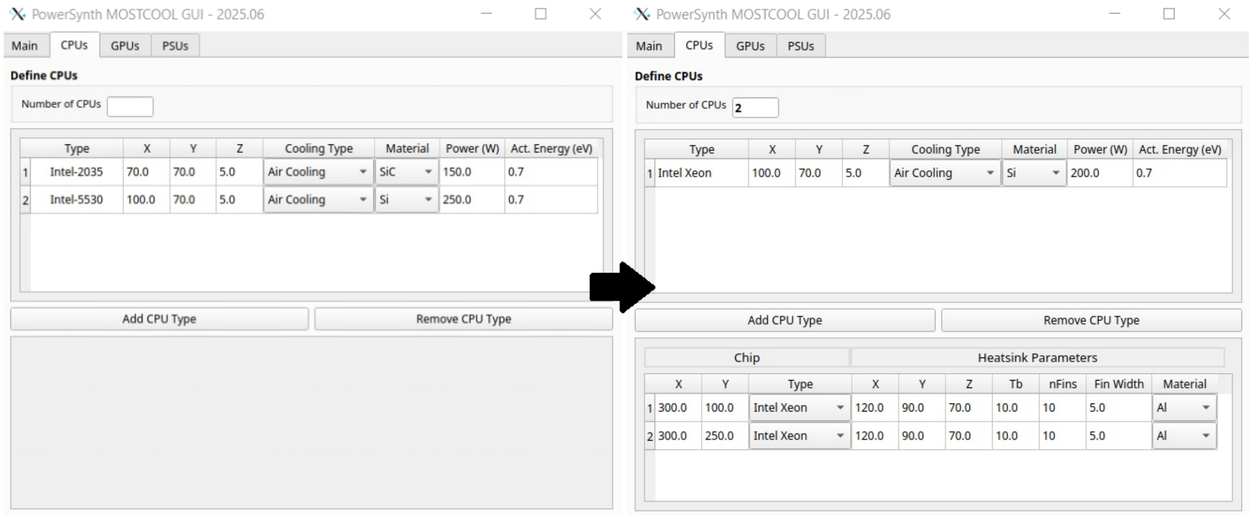


Figure 2: PowerSynth2-MOSTCOOL CPUs tab: Initial and Final setting up parameters

(c) **GPUs:** In this tab users should define GPUs parameters. The parameters are the same as the CPUs tab. (shown in Figure 3)

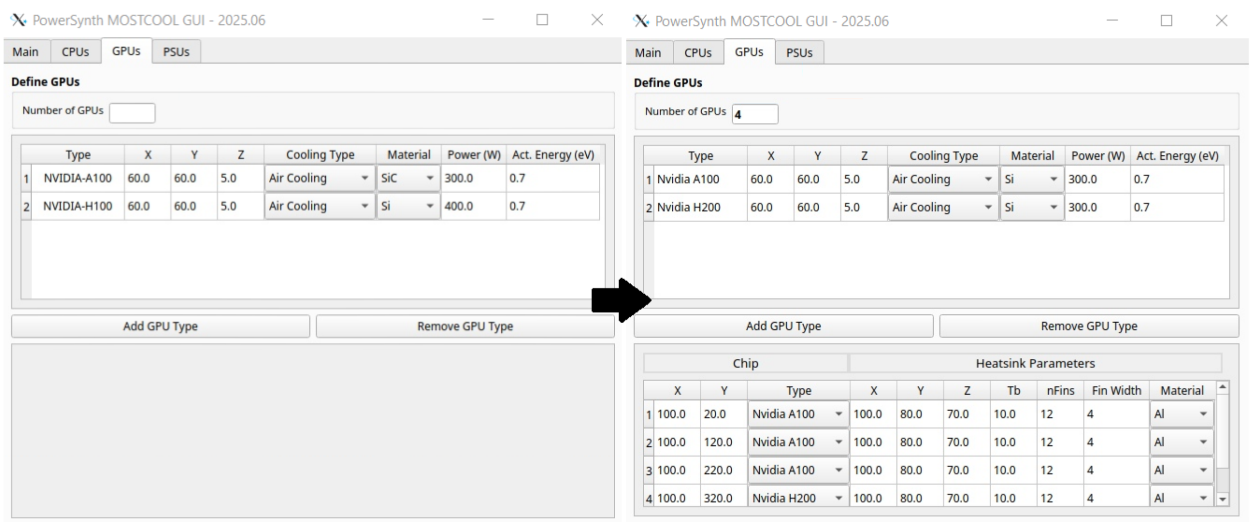


Figure 3: PowerSynth2-MOSTCOOL GPUs tab: Initial and Final setting up parameters

(d) **PSUs:** In this tab users should define PSUs parameters. The parameters are the same as the CPUs tab except for PSUs do not have heatsinks. (shown in Figure 4)

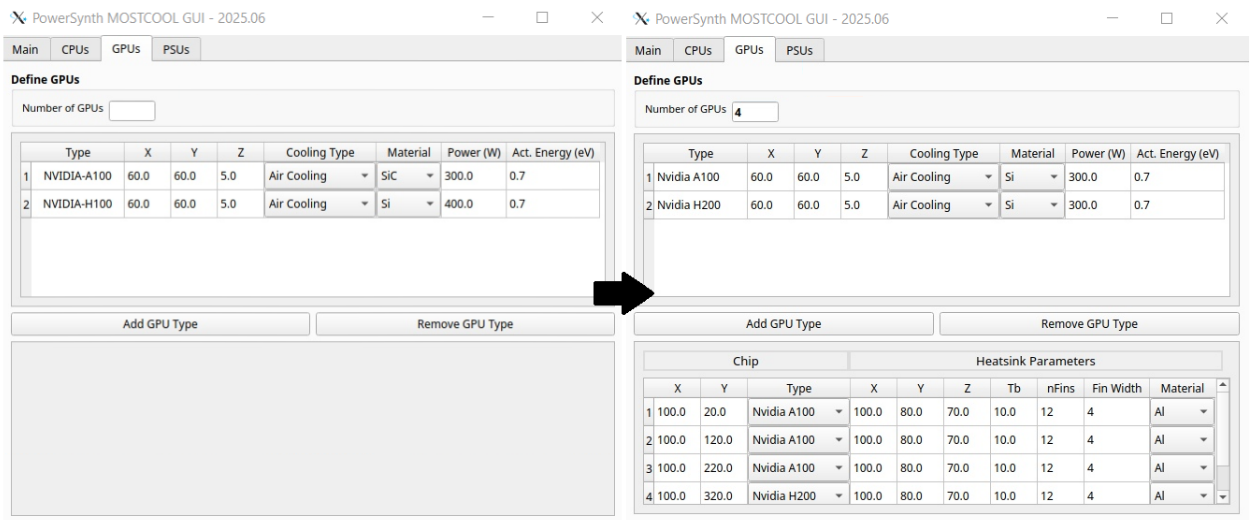


Figure 4: PowerSynth2-MOSTCOOL PSus tab: Initial and Final setting up parameters

After run the simulation, the Solution Browser window will appear. The window has three tabs. (a) 2D plot tab to represent interactive 2D plots of different layers, 3D PNG tab to show 3D view of the server's temperature map, and summary tab includes the summary of the results as a table.

(a) **2D Plot:** This tab shows different layers of server's temperature map. the user can zoom, save the figure in different format. Also, by clicking on the server board, the x,y and corresponding temperature will show in the toolbar. The user can toggle between different layers; components layer that includes, CPUs, GPUs, PSUs and PCB layer. Moreover, the user can change the range of temperature for more analysis. Figure 5 is shown the 2D Plot tab.

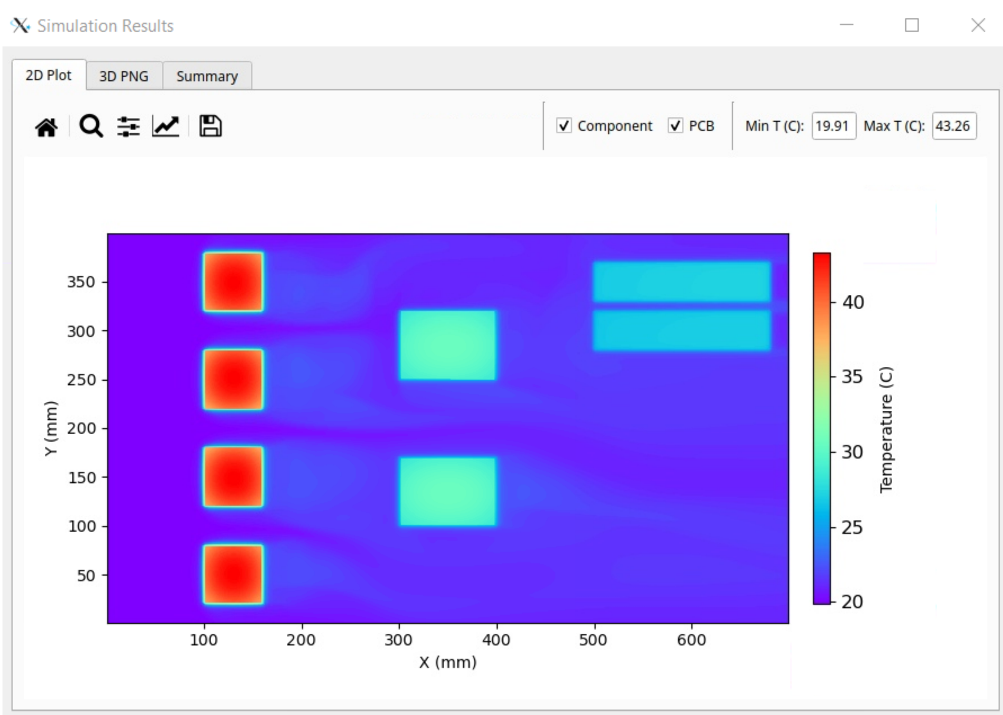


Figure 5: PowerSynth2-MOSTCOOL solution window: 2D Plot tab

(b) **3D PNG:** This tab shows the static 3D of server's temperature map. Figure 6 is shown the 3D Plot tab.

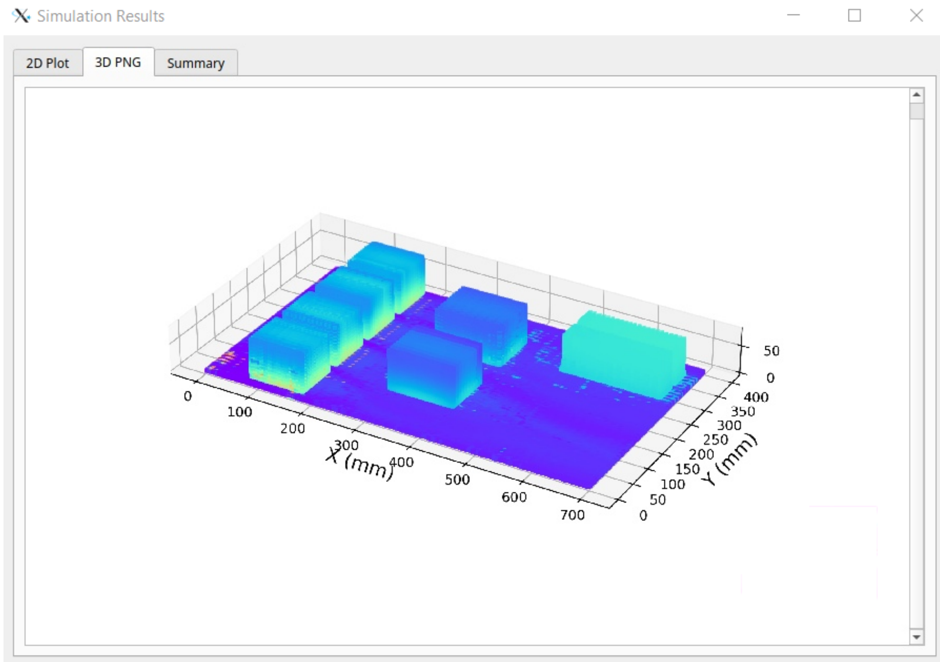
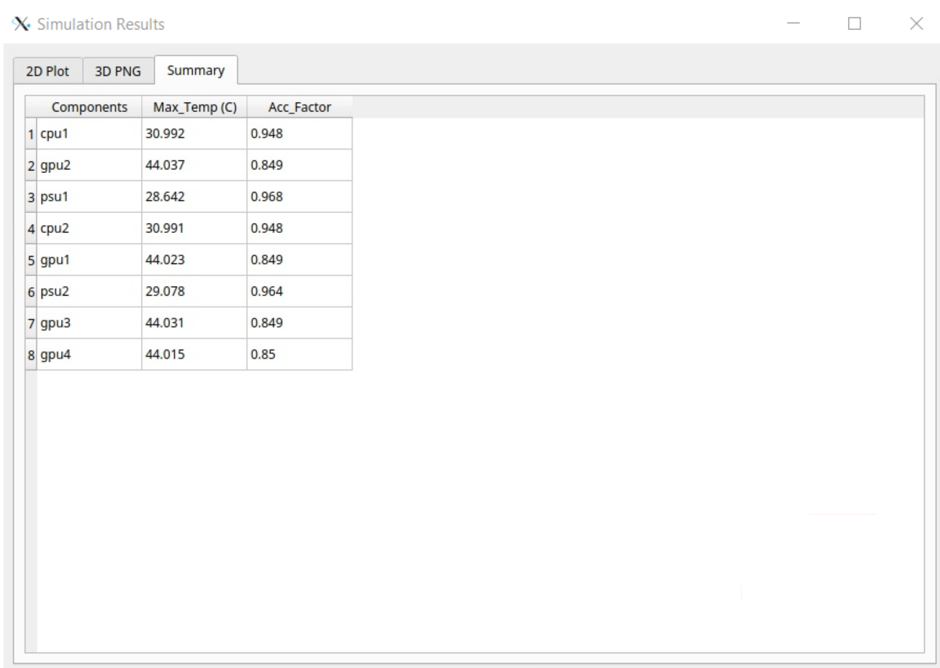


Figure 6: PowerSynth2-MOSTCOOL solution window: 3D PNG tab

(c) **Summary:** This tab shows a table which consists of components, maximum temperature, and reliability index. Figure 7 is shown Summary tab.



	Components	Max_Temp (C)	Acc_Factor
1	cpu1	30.992	0.948
2	gpu2	44.037	0.849
3	psu1	28.642	0.968
4	cpu2	30.991	0.948
5	gpu1	44.023	0.849
6	psu2	29.078	0.964
7	gpu3	44.031	0.849
8	gpu4	44.015	0.85

Figure 7: PowerSynth2-MOSTCOOL solution window: Summary tab

3 Walk-Through Examples

In the Sample Design we consider two examples: 1 unit server which consist of 2 CPUs, 2 GPUs, and 2 PSUs, and 2 unit server which consist of 2 CPUs, 4 GPUs, and 2 PSUs. In this section, 2 unit server is demonstrated step-by-step.

3.1 2 Unit Server

The user can define the server by the UI. Also, more convenient way is that define the server in YAML file. Listing 1 is shown the YAML file of this example.

Listing 1: 2UServer YAML file

```yaml
case_study:
  name: "S2U2CPU4GPU"
solver_settings:
  iteration: 100
  number_of_cores: 16
reliability:
  mode: 1  # 0=no reliability, 1=simple reliability, 2=advanced
     reliability
  test_temperature: 298.15  # in Kelvin
boundary_conditions:
  inlet_temperature: 293.15 # in Kelvin
  air_velocity: 10.0        # in m/s
  air_direction: "LR"       # Options: LR, RL, FB, BF
  tim_thickness: 0.0005     # in mm
  tim_kappa: 10.0           # in W/m2K
  unit: "mm"                # mm, cm, m
define_server:
  pcb:
    x: 700  # in mm
    y: 400  # in mm
    z: 3    # in mm
    material: "FR4"
  cabinet:
    x: 720  # in mm
    y: 420  # in mm
    z: 90   # in mm
define_components:
  number_of_cpus: 2
  number_of_gpus: 4
  number_of_psus: 2
cpus:
  - id: "Intel Xeon"
    x_location: 300.0  # X Location in mm
    y_location: 100.0  # Y Location in mm
    type: "Air Cooling"  # Options: "Air Cooling", "Water Cooling"
    x_dim: 100.0  # X Dimension in mm
```

```
    y_dim: 70.0   # Y Dimension in mm
    z_dim: 5.0    # Z Dimension in mm
    material: "Si"
    power: 200.0  # Power in Watts
    activation_energy: 0.7  # Activation Energy
    heatsink:
      checked: true  # Whether heatsink is enabled
      x: 120.0  # Heatsink X in mm
      y: 90.0   # Heatsink Y in mm
      z: 70.0   # Heatsink Z in mm
      tb: 10.0  # Tb parameter
      n_fins: 10  # Number of fins
      fin_width: 5.0  # Fin width in mm
      material: "Al"
...
gpus:
  - id: "Nvidia A100"
    x_location: 100.0
    y_location: 20.0
    type: "Air Cooling"
    x_dim: 60.0
    y_dim: 60.0
    z_dim: 5.0
    material: "Si"
    power: 300.0
    activation_energy: 0.7
    heatsink:
      checked: true
      x: 100.0
      y: 80.0
      z: 70.0
      tb: 10.0
      n_fins: 12
      fin_width: 4
      material: "Al"
...
psus:
  - id: "PSU 100"
    x_location: 500.0  # X Location in mm
    y_location: 280.0  # Y Location in mm
    x_dim: 180.0     # X Dimension in mm
    y_dim: 40.0     # Y Dimension in mm
    z_dim: 70.0     # Z Dimension in mm
    material: "copper"  # Material
    power: 100.0    # Power in Watts
    activation_energy: 0.7  # Activation Energy
```

In Listing 1 first section is case study and has a name as a variable. The solver_settings consist of iteration and number of cores for simulation. In reliability section users can define the mode of reliability and test temperature for reliability calculation. In boundary_condition users can define OpenFOAM BCs. In define_server the dimension of PCB and cabinet should be define. In define_components the number of CPUs, GPUs, and PSUs should be define by users.

The next section is related to CPUs. every CPU should be define. It consist of CPU's dimensions, power, material, heatsink parameters, and the location on the board. The GPUs and PSUs section are the same as CPUs.

Now, run PowerSynth2-MOSTCOOL executable by running the PS2MostCool command. Then, follow the GUI instructions and provide necessary files according to the figures shown below. After the main window appears, the user should press the load button. (shown in Figure 8)

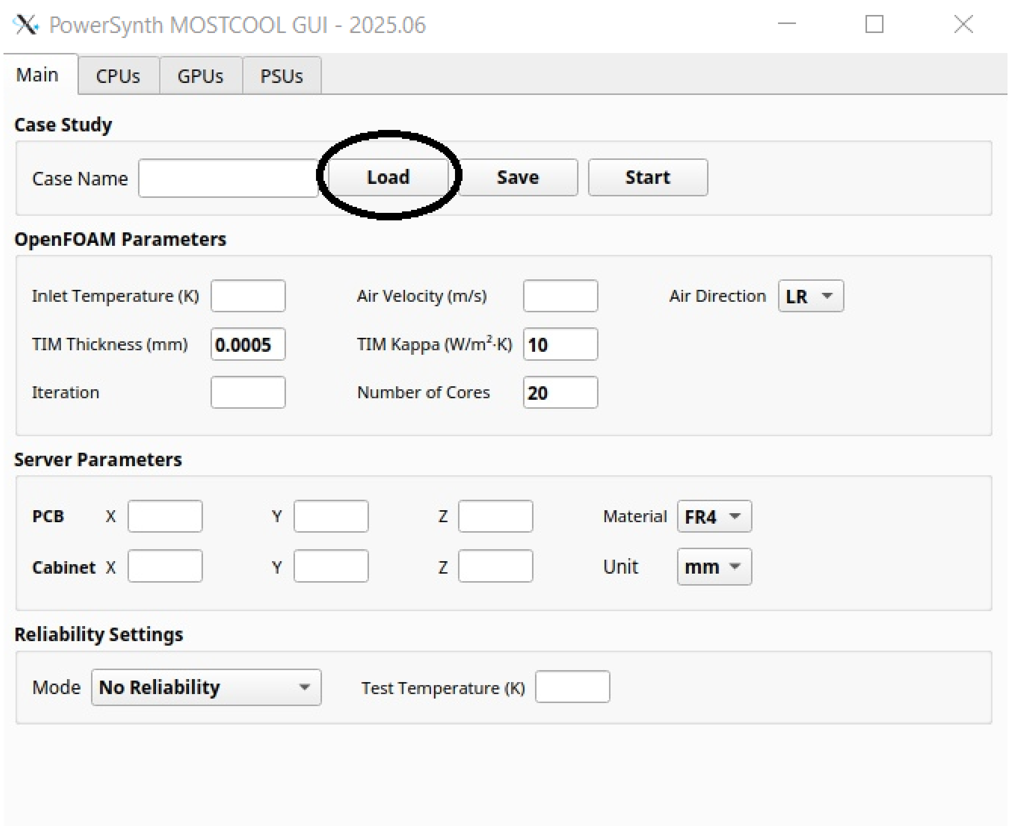


Figure 8: PowerSynth2-MOSTCOOL Main window: Main tab

For now, we consider two examples. In the opening window, users can see both of them. We are going to the 2Userver example. Then, in the folder select the YAML file and press the open button. (shown in Figure 9)

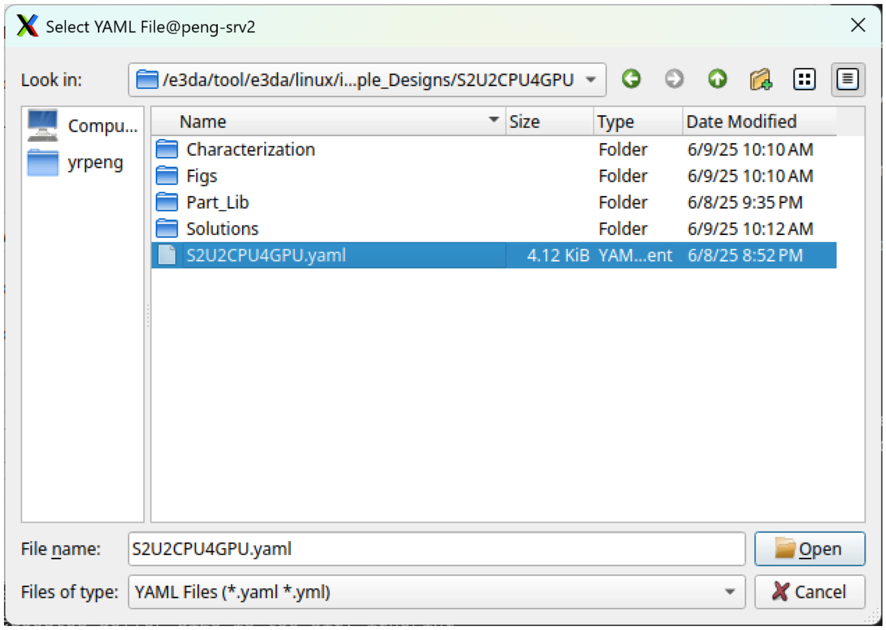


Figure 9: PowerSynth2-MOSTCOOL load case.

In Figure 10 you can see that all of data loaded on the UI. Users can change the values from UI. for example they can change the power of components, or the heatsink parameters. It is noted that any changes on the UI do not change the YAML file. If the user want to have the changes they should save a new YAML using Save button in Main tab.

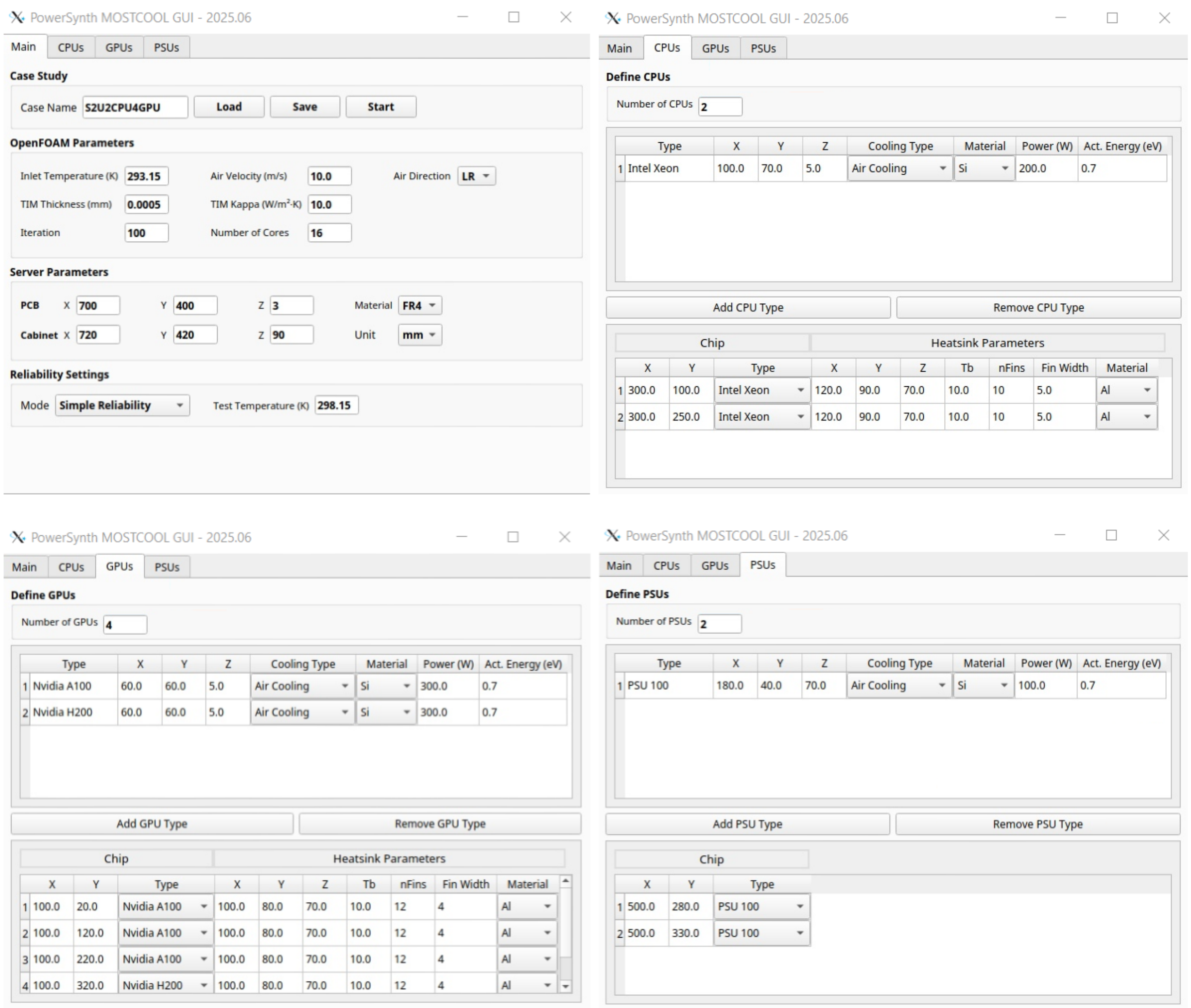


Figure 10: PowerSynth2-MOSTCOOL UI.

Now, by pressing the Start button the simulation will begin. After finishing the simulation the solution window will appear. (shown in Figure 11)

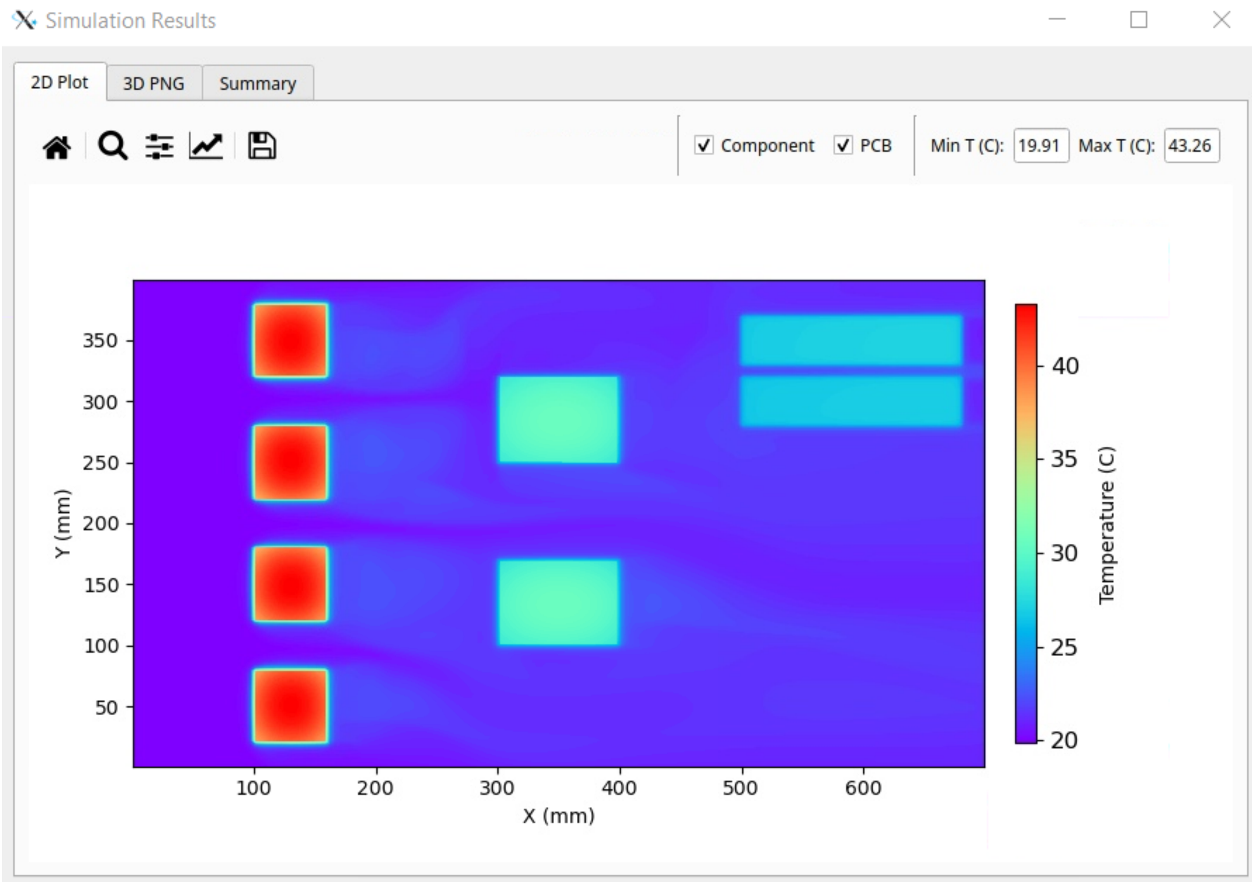
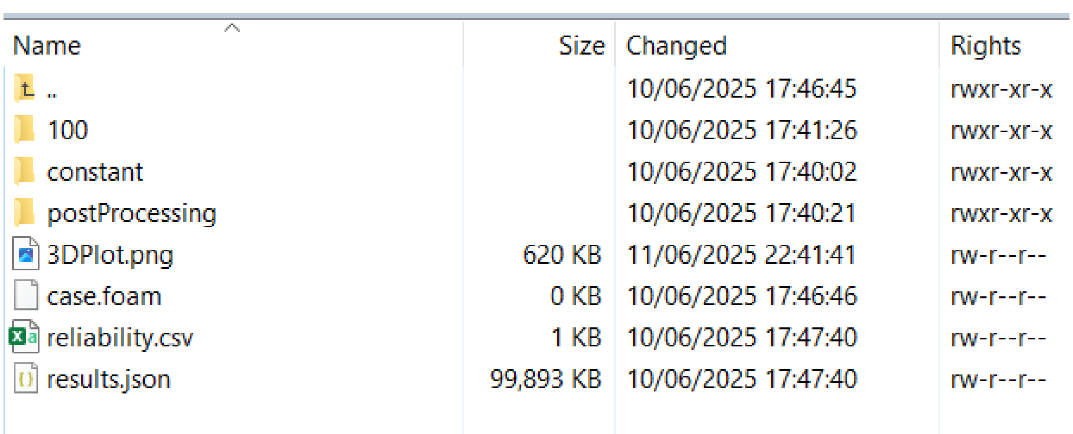


Figure 11: PowerSynth2-MOSTCOOL Solution window.

The users can access the results in Solution folder.

./pkg/work/Sample_Designs/S2U2CPU4GPU/Solutions/Initial_Layout/Layout0. Results folder is shown in Figure 12 consists of the OpneFOAM case with the last iteration of the simulation. Also, the reliability.csv and results.json are exist for postprocessing. The user can use the ParaView software for postprocessing. It is noted that the users need to first install ParaView.



Name	Size	Changed	Rights
..		10/06/2025 17:46:45	rwxr-xr-x
100		10/06/2025 17:41:26	rwxr-xr-x
constant		10/06/2025 17:40:02	rwxr-xr-x
postProcessing		10/06/2025 17:40:21	rwxr-xr-x
3DPlot.png	620 KB	11/06/2025 22:41:41	rw-r--r--
case.foam	0 KB	10/06/2025 17:46:46	rw-r--r--
reliability.csv	1 KB	10/06/2025 17:47:40	rw-r--r--
results.json	99,893 KB	10/06/2025 17:47:40	rw-r--r--

Figure 12: PowerSynth2-MOSTCOOL Results Folder.

4 Useful Links

- **Release Website:** https://e3da.csce.uark.edu/release/PowerSynth/
- **Publication Website:** https://e3da.csce.uark.edu/pub/
- **OpenFOAM Website:** https://www.openfoam.com/news/main-news
- **ParaView Website:** https://www.paraview.org/download/